MODULE 2 · STATISTICS & PROBABILITY · TOPIC 2.9

Correlation

Day 29 — one number for "do these move together?", and four ways it misleads you

1	4	4
Number, measuring one thing only	Pictures it cannot tell apart	Reasons for a correlation, one of which is cause

Prepared & presented by **Srinivasa Sai Chava**

Boston University

WHY THIS SESSION

Everyone knows the slogan. Almost nobody knows the four reasons.

“Correlation is not causation”

True, and almost useless on its own. It tells you what the answer is NOT. It does not tell you what else could be going on, so people repeat it and then quietly believe the causal story anyway.

The useful version

If X and Y are correlated, there are exactly four explanations, and you should be able to name all four and argue about which applies.

By the end of today you will be able to. Only one of the four is “X causes Y”.

Four things that will happen today

1. Four completely different pictures will produce the identical r = 0.816
2. One extra data point will turn r = −0.14 into r = +0.64, with p = 0.00000000000047
3. Two variables with zero relationship will correlate at −0.73
4. Women will be admitted more often in every department, and less often overall

WARM-UP · 5 MINUTES

Four questions from Day 28

1	**What is a p-value, exactly?**	→ P(data this extreme | nothing is happening).
2	**Which scipy default is dangerous?**	→ equal_var=True. Use Welch.
3	**20 tests on noise — chance of a "finding"?**	→ 64%.
4	**What should you report besides p?**	→ Effect size, interval, n, and how many tests.

Hold on to question 3. A dataframe with 20 numeric columns contains 190 possible correlations, and today you will watch someone "discover" one in data that has nothing in it.

SESSION PLAN

How today's two hours are structured

1. **What r measures** — 15 min
   And the two things it does not

2. **What r cannot see** — 25 min
   Curves, outliers, and Anscombe

3. **How well do you know r?** — 15 min
   The interval, and tiny r at big n

4. **The four explanations** — 25 min
   Only one of them is cause

5. **Searching for correlations** — 10 min
   190 pairs and nothing real

6. **Practice notebook** — 30 min
   You code, I walk around

Section 4 is the one worth the whole session. It is the difference between repeating a slogan and being able to argue about a causal claim.

1

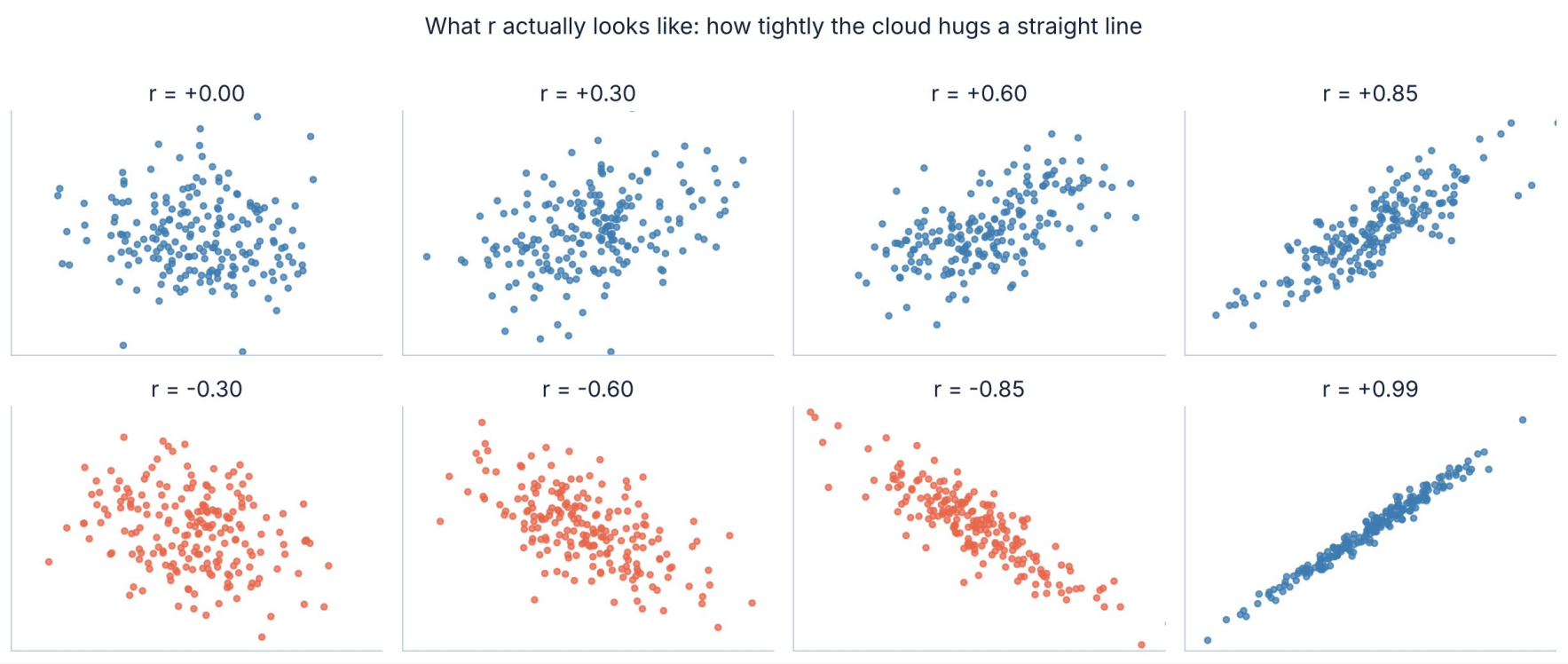
What r Measures

One number, one job · 15 minutes

- What the values look like
- r is not a slope
- r squared

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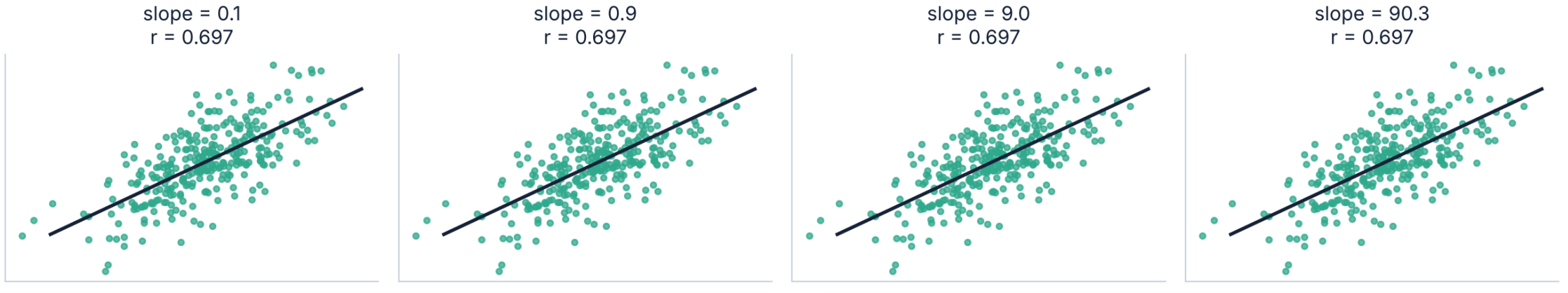
What the numbers actually look like



Pearson's r runs from −1 to +1. The sign says which way the cloud leans; the size says how tightly it hugs a straight line. r = 0 is a shapeless blob and r = ±1 is a perfect line. Note that r = 0.3 — which sounds like “some relationship” — looks like almost nothing.

⚠️ r tells you nothing about how big the effect is

Four completely different slopes. Identical r. r measures tightness, not steepness.



Same r, slopes spanning 1000×

All four panels have r = 0.697. One of them gains 0.09 units of y per unit of x; another gains 90. r cannot tell them apart, because it is computed after both variables are standardised.

So what does r mean?

Only: how predictable y is from x. Not how much y changes. If someone asks “how much does revenue rise per extra salesperson?”, r is the wrong number — you want the slope, which is Module 3.

r^2 — the number that keeps you honest

Square it, and it becomes meaningful

r^2 is the share of the variation in y that x accounts for. It is the number to quote when somebody sounds impressed by a correlation.

r = 0.5 sounds like half a relationship.
It accounts for 25% of the variation,
which leaves 75% unexplained.

r	r^2	reads as
0.10	1%	essentially nothing
0.30	9%	weak
0.50	25%	moderate
0.70	49%	strong
0.90	81%	very strong

The honest habit

Whenever you report an r, report r^2 next to it. “Hours studied correlates with marks at r = 0.3” sounds like a finding worth acting on. “Hours studied accounts for 9% of the variation in marks” is the same fact, and it is the version that stops anyone redesigning a syllabus around it.

These adjectives are conventions, not laws — in physics r = 0.9 is disappointing, and in psychology r = 0.3 is a strong result. What counts as strong depends entirely on the field.

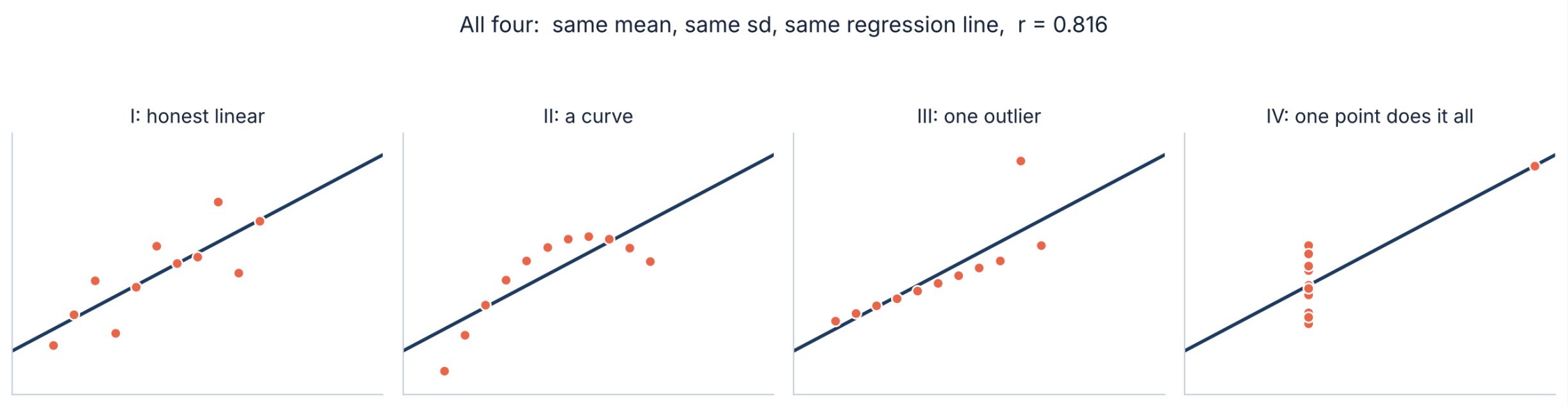
2

What r Cannot See

Four pictures, one number · 25 minutes

- Anscombe's quartet
- Curves
- Ranks
- One bad point

⭐ Four datasets. Every summary statistic identical.



What is the same in all four

Mean of x = 9.00. Mean of y = 7.50. sd of y = 2.03. Regression line y = 3 + 0.5x. And r = 0.816 in all four (they differ in the fourth decimal place).

Only one of them is a relationship you would want to model. Francis Anscombe built these in 1973 to make a single point: ALWAYS PLOT THE DATA. Every number you could compute agrees, and the pictures do not.

r only ever asks about straight lines

In every one of these, x pins y down to one or two values. Pearson's r reports nothing.



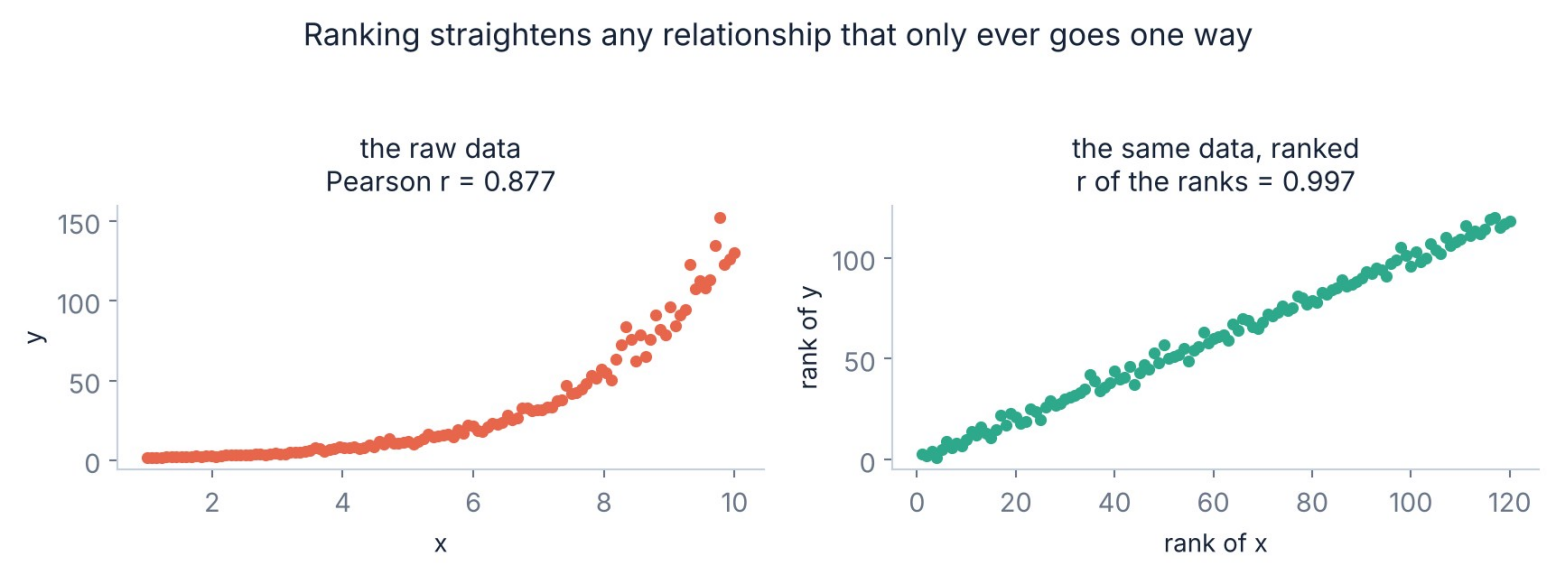
All four are near-perfect relationships

In every panel, x pins y down to one or two values. Every r comes back under 0.07, because r asks one narrow question: as x goes up, does y go up in a STRAIGHT LINE?

Where you meet this for real

Dose and effect (too little does nothing, too much harms). Price and demand. Practice and performance. Age and almost anything. U-shapes are everywhere, and r = 0.01 on a U-shape reads as “no relationship”.

Spearman: the fix when the curve only goes one way



Spearman

= Pearson's r, computed on the RANKS instead of the values.

It asks: when x goes up, does y go up?

Not: does it go up in a straight line?

```
stats.spearmanr(x, y)
```

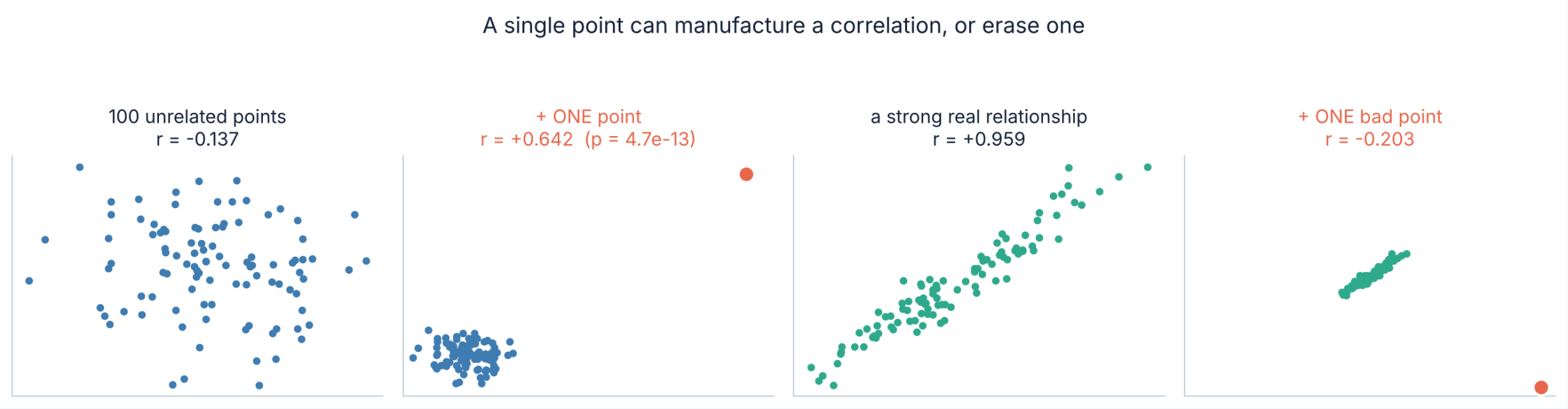
What ranking does

Replace every value by its position in the order. Pearson saw 0.877 on the curve; r on the ranks comes out at 0.997 — the relationship is essentially perfectly monotonic, and ranking is what reveals that.

When to reach for it

A curved but one-directional relationship · heavy outliers · ordinal data like star ratings. But Spearman still cannot see a U-shape: it goes up then down, so the ranks do not line up either.

⚠️ One point can manufacture a correlation, or erase one



Left pair: inventing a relationship

100 unrelated points give r = −0.14. Add one point at (200, 200) — a typo, a different unit, a test record — and r becomes +0.64 with p = 0.00000000000047.

Right pair, and the fix

A genuine r = 0.96 collapses to −0.20 from one bad point. Spearman barely moves in either case, because a rank cannot run away. Plot first; compare Pearson and Spearman second.

3

How Well Do You Know r?

The number has error bars too · 15 minutes

- The interval for r
- Tiny r at large n

r is an estimate, and estimates have intervals



r = 0.50 from 10 points

is compatible with the true correlation being anywhere from −0.19 to +0.86. It includes zero, and it includes almost perfect.

You have learned essentially nothing, and the number 0.50 hides that completely.

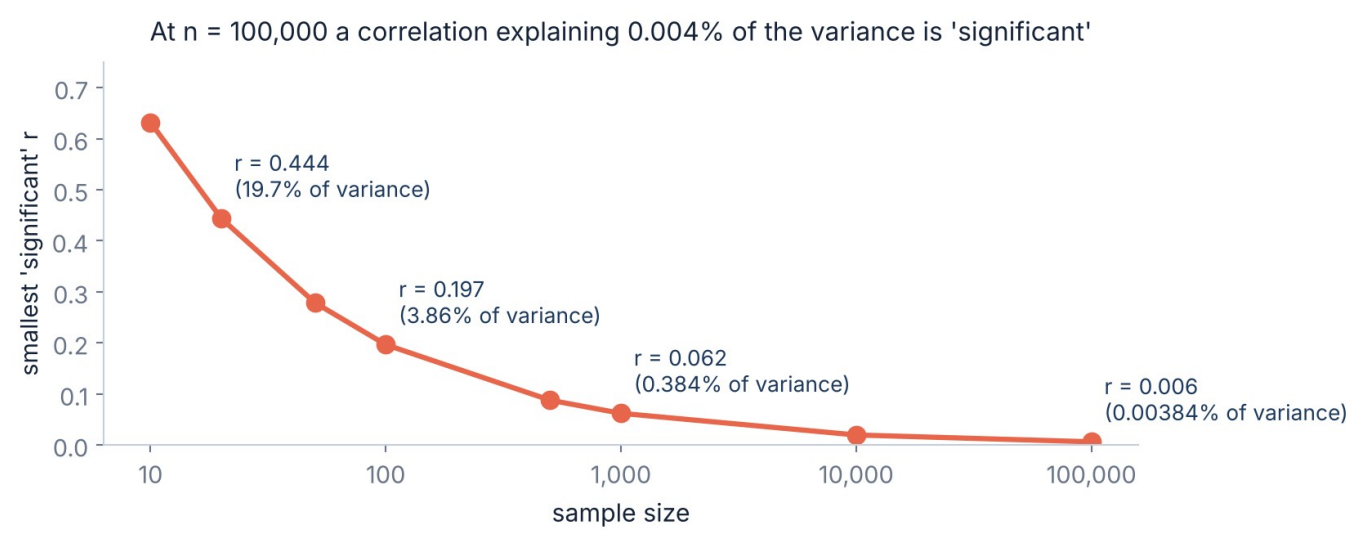
Correlations from small samples are far less stable than they look. Below about n = 30, treat any r as a hint rather than a measurement — and always print the interval beside it.

How to get it

```
res = stats.pearsonr(x, y)
res.confidence_interval()
```

Built into scipy. One extra line, and it stops you over-reading a small sample.

⚠️ At large n, a meaningless r is “significant”



Note the two ends of this section together: with too little data r is unstable, and with a great deal of data its p-value stops meaning anything. Neither problem is solved by looking harder at r itself.

The same trap as yesterday

With 100,000 rows, any correlation above 0.006 reaches $p < 0.05$. That is an r^2 of 0.004% — a relationship so faint it is indistinguishable from nothing.

The p-value says “real”. The r^2 says “irrelevant”. Both are right.

Which is why r^2 goes next to p

In a dataset of any real size, “statistically significant correlation” carries almost no information. The question that matters is how much of the variation it accounts for — and that is r^2.

4

The Four Explanations

Only one of them is cause · 25 minutes

- Chance
- Confounding
- Reverse causation
- Selection

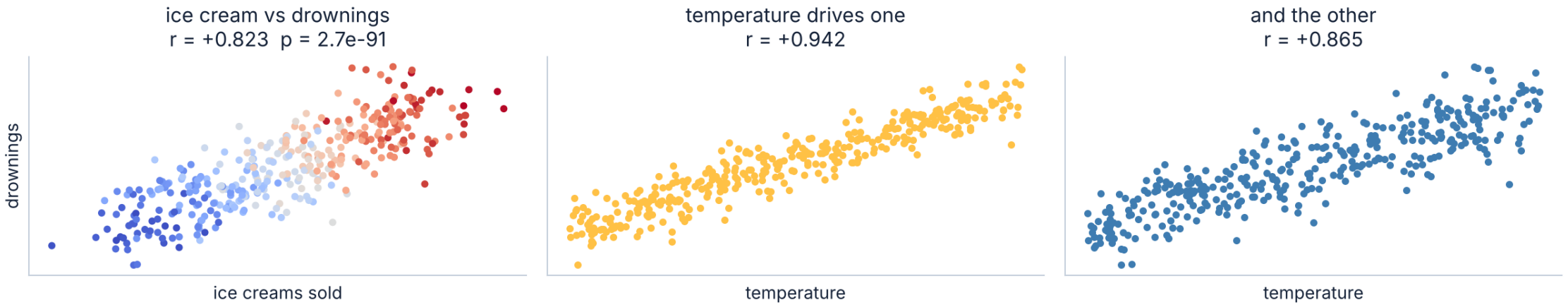
X and Y are correlated. There are four reasons.

1	**Chance**	You looked at enough pairs that one came up. Section 5.
2	**Confounding**	Something else causes both. Ice cream and drownings.
3	**Reverse causation**	Y causes X, not the other way round.
4	**Selection**	The way the data was collected created the link.
5	**X really does cause Y**	The one everybody assumes. Last on the list for a reason.

Five, not four — because "X causes Y" is one item on the list, not the default. When you meet a correlation, your job is to argue about which of these it is.

Confounding: something else causes both

Colour is temperature. Once you know it, the first panel has nothing left (partial r = 0.05).



$r = 0.82$, $p = 3 \times 10^{-91}$

That is about as strong and as significant as correlations get in real data. Ice cream does not drown anyone. Hot days sell ice cream and hot days put people in the water.

What to do about it

Hold the confounder still. Compare ice cream and drownings WITHIN days of the same temperature and the correlation drops from 0.82 to 0.05. That is a partial correlation, and it is Module 3's main business.

Reverse causation: the arrow points the other way

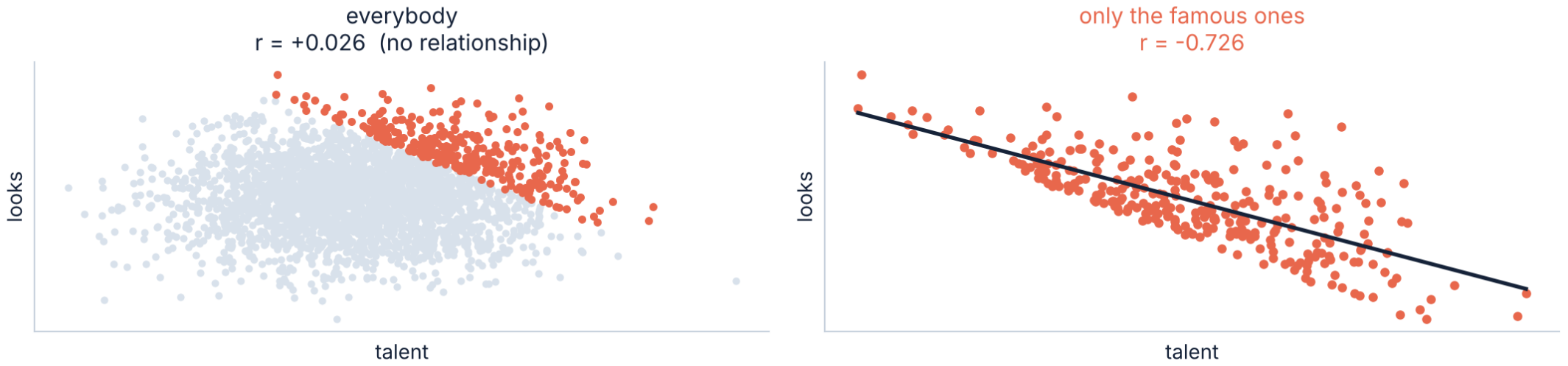
A correlation is symmetric. r(x, y) = r(y, x), exactly. The data cannot tell you which way the arrow points — that information is simply not in it.

“Hospitals with more staff have more deaths”	→ Bigger hospitals take the sicker patients. Severity drives both.
“People who diet weigh more”	→ Weighing more is why they started dieting.
“Firms using more consultants perform worse”	→ Struggling firms hire consultants.
“Heavier police presence, more crime”	→ Crime draws police to the area.

The test: ask whether the story works just as well backwards. If it does, your data cannot distinguish them, and you need either a time ordering or an experiment.

⭐ Selection: when collecting the data creates the link

You get famous with enough of EITHER. Among the famous, they trade off — and nothing caused anything.



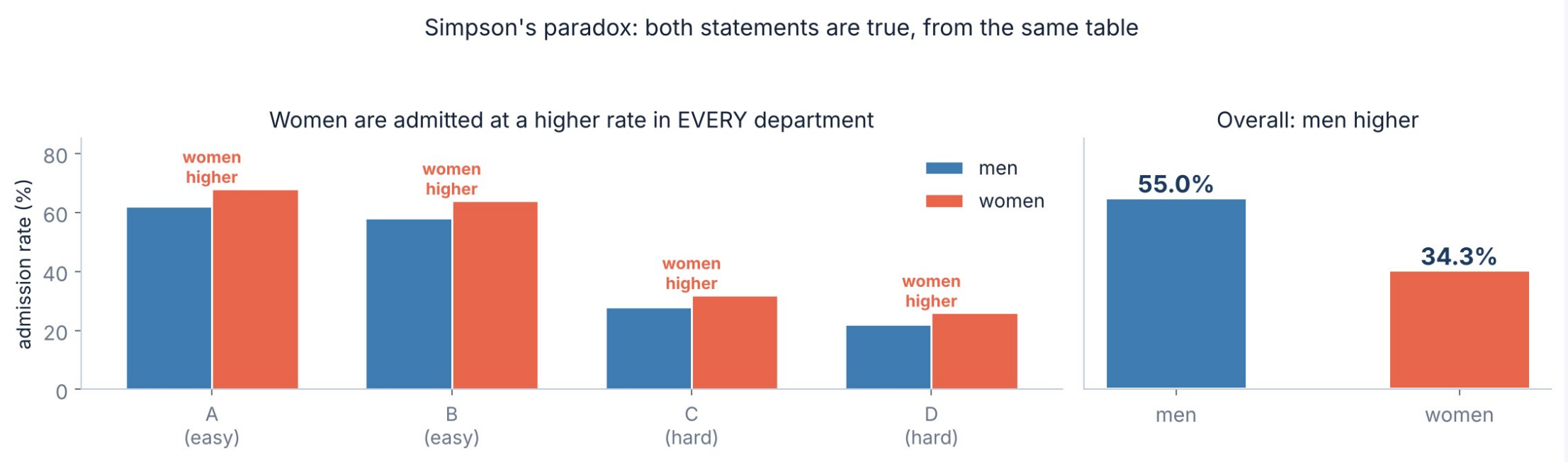
Nobody caused anything

Talent and looks are independent in the population. Among famous people they correlate at −0.73, because you need enough of one or the other to get in.

Where it bites you

Any analysis on people who were admitted, hired, diagnosed, subscribed or survived. The selection rule itself writes correlations into the data.

⚠️ Simpson's paradox: the reversal



Both statements are true

Women are admitted at a higher rate in every single department, and at a lower rate overall (34.3% vs 55.0%). Both come from the same table, with no arithmetic error anywhere.

Why: where people applied

85% of the men applied to the easy departments; only 14% of the women did. The overall rate mostly measures which doors people knocked on, not what happened at the door.

What to ask before believing any correlation

1	**Did I plot it?**	Anscombe. Non-negotiable, and it takes four seconds.
2	**How many pairs did I look at?**	If it was 190, this one means nothing on its own.
3	**What is r^2, and the interval?**	Not just r. Both ends: is it big, and do I know it?
4	**What else could cause both?**	Name a specific confounder, then try to control for it.
5	**Does the story work backwards?**	If yes, the data cannot tell you which way it goes.
6	**Who is in this dataset, and why?**	Selection. Admitted, hired, survived, subscribed.

Questions 4, 5 and 6 cannot be answered from the data. They need you to know where the numbers came from — which is why a domain expert beats a bigger dataset.

5

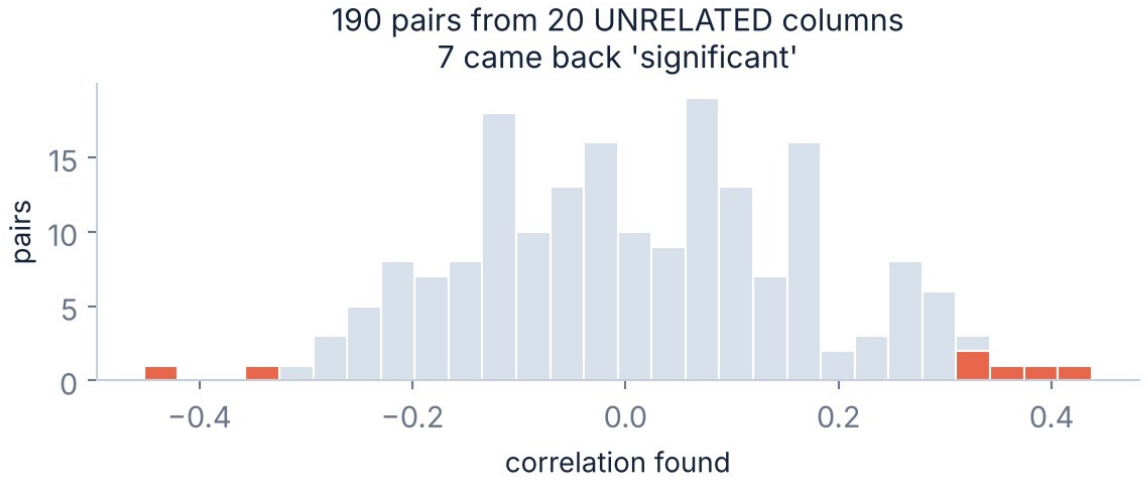
Searching for Correlations

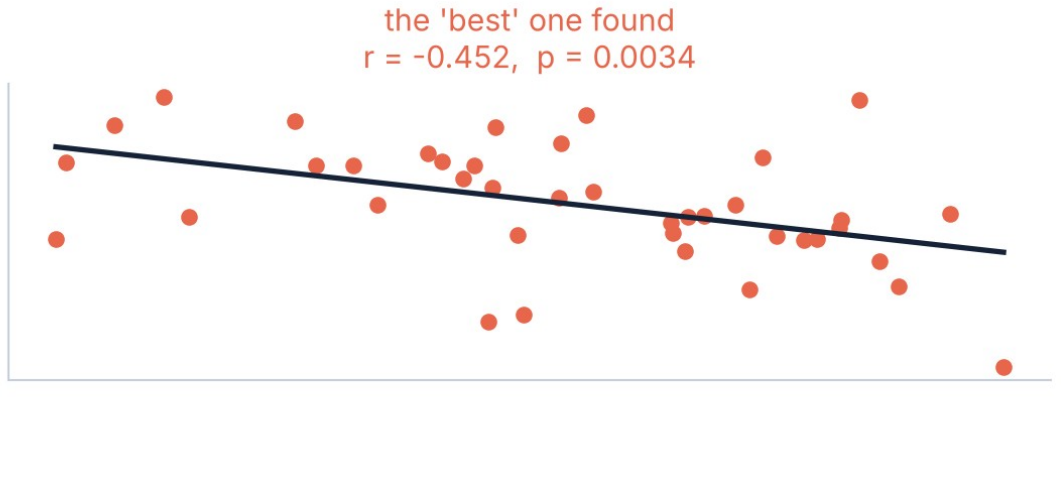
Yesterday's lesson, today's costume · 10 minutes

- 190 pairs
- What to report

⚠️ df.corr() on 20 columns runs 190 tests

There is nothing in this data. Searching hard enough found something anyway.





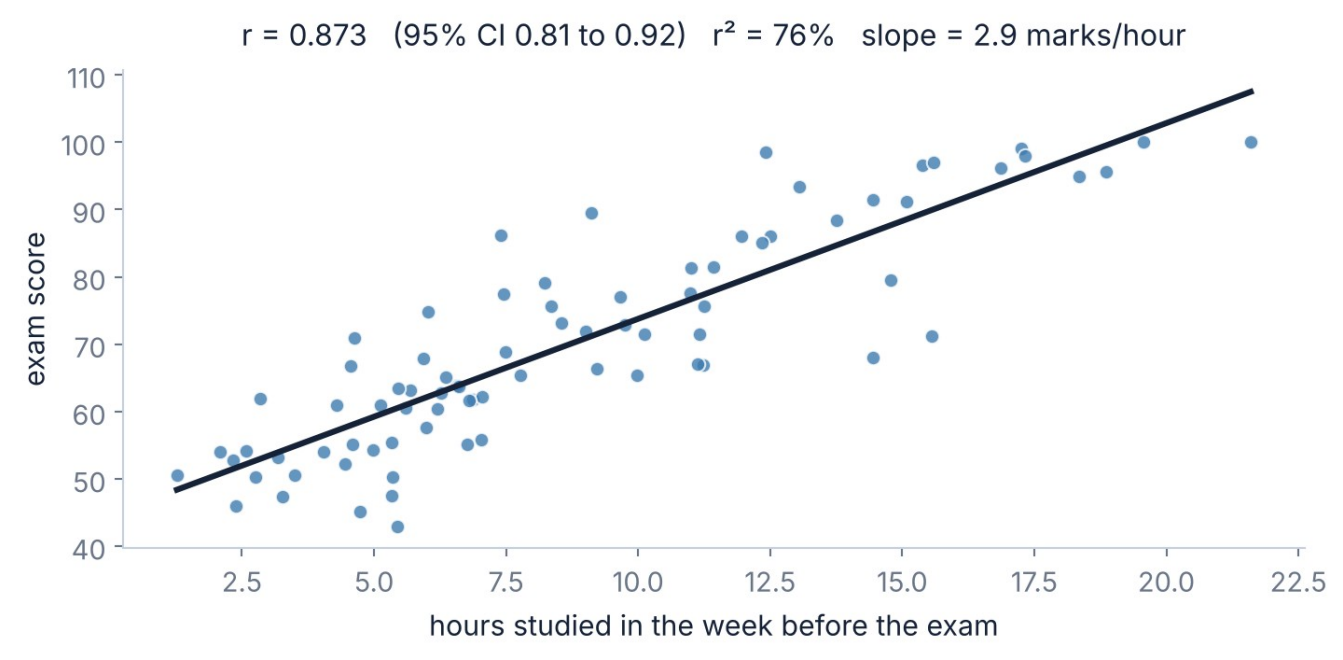
There is nothing in this data

20 independent random columns, 40 rows. Seven pairs came back $p < 0.05$, and the strongest reached $r = 0.45$ with $p = 0.003$. It looks like a finding.

The honest handling

The Bonferroni threshold for 190 pairs is 0.00026. Nothing survives it, correctly. A heatmap is a tool for generating hypotheses — never for confirming one.

Worked through: study hours and exam scores



The sentence to write: "Hours studied and exam score are strongly related (r = 0.87, 95% CI 0.81–0.92, n = 80; r^2 = 0.76). This is observational, so we cannot separate studying from the motivation that drives it."

What we can say

r = 0.873, 95% CI 0.81 to 0.92, n = 80. Spearman agrees at 0.883, so it is not an outlier artefact. r^2 = 76% of the variation in scores.

The slope says about 2.9 marks per extra hour.

⚠️ What we cannot say

That studying causes the marks. Motivation plausibly drives both (confounding). Students who find it easy may study more because it is rewarding (reverse). And this is one class — who dropped out before the exam?

Which correlation, and what to report

Situation	Use	scipy
Straight-line relationship, no wild outliers	Pearson	`stats.pearsonr(x, y)`
Curved but always one direction	Spearman	`stats.spearmanr(x, y)`
Heavy outliers you cannot remove	Spearman	`stats.spearmanr(x, y)`
Ordinal data (star ratings, grades)	Spearman or Kendall	`stats.kendalltau(x, y)`
Small sample with many ties	Kendall's tau	`stats.kendalltau(x, y)`
U-shaped or non-monotonic	none of them — plot it	`plt.scatter(x, y)`
Need the size of the effect	regression slope	`np.polyfit(x, y, 1)`

Always report together: r, its 95% confidence interval, r^2, n, and whether you checked Spearman. Five numbers, one line of output, and a reader can judge everything that matters.

RECAP

Debug corner — eight correlation mistakes

You did	What goes wrong	Fix
computed r without plotting	Anscombe: four pictures, one r	plot it, every time
read r as the size of the effect	slopes 0.09 and 90 share r = 0.697	report the slope too
reported r = 0.3 as a relationship	that is 9% of the variation	report r^2 beside it
used Pearson on a U-shape	r ≈ 0 on a perfect relationship	plot it; Spearman will not help either
left one typo in the data	r went from −0.14 to +0.64	compare Pearson with Spearman
computed r on 10 rows	95% CI was −0.19 to +0.86	report the interval
scanned df.corr() for the biggest	190 silent tests	Bonferroni, or pre-specify the pair
concluded X causes Y	four other explanations exist	work the checklist

Row 1 prevents rows 2, 4 and 5 on its own. If your students only change one habit this session, make it that one.

Putting it together — a correlation you can trust

```python
import numpy as np
import matplotlib.pyplot as plt
from scipy import stats

def correlate(x, y, n_pairs=1, labels=("x", "y")):
    """Report a correlation with everything needed to judge it."""
    x, y = np.asarray(x, float), np.asarray(y, float)

    #  1. always plot. This single line prevents most mistakes.
    plt.scatter(x, y, s=18, alpha=.7)
    plt.xlabel(labels[0]); plt.ylabel(labels[1]); plt.show()

    #  2. both coefficients - a gap between them means outliers
    pe = stats.pearsonr(x, y)
    sp = stats.spearmanr(x, y)
    lo, hi = pe.confidence_interval()

    print(f"n = {len(x)}")
    print(f"Pearson  r = {pe.statistic:+.3f}  95% CI [{lo:+.3f}, {hi:+.3f}]")
    print(f"Spearman r = {sp.statistic:+.3f}")
    print(f"r^2 = {pe.statistic**2:.1%} of the variance")
    print(f"slope = {np.polyfit(x, y, 1)[0]:+.3f} {labels[1]} per {labels[0]}")
    print(f"p = {pe.pvalue:.4f}   threshold {0.05/n_pairs:.5f} "
          f"({n_pairs} pair(s) examined)")
    if abs(pe.statistic - sp.statistic) > 0.15:
        print("  !! Pearson and Spearman disagree - check for outliers")
```

It plots before it prints

the habit that catches Anscombe

Both coefficients, always

a gap between them IS the warning

The interval, not just r

n = 10 is a hint, not a number

r^2 next to r

9% sounds different to 0.3

The slope, for size

r cannot answer “how much?”

n_pairs is a required thought

df.corr() is 190 tests

What it cannot do is tell you whether X causes Y. No function can. That is the checklist, and it needs you.

OVER TO YOU

Practice session — 30 minutes

A **Predict the output** — 8 questions

r vs slope, curves, ranks, outliers

B **Spot & fix the bug** — 4 questions

The unplotted r, the matrix scan, a causal leap

C **Write the code** — 5 tasks

Diagnose, control, and report

D **Challenge** — 2 tasks

Build a fake finding, then break a real one

Your notebook

`Day29_Practice_Questions.ipynb`

- Plot it before you compute it
- Pearson AND Spearman
- How many pairs did I look at?
- Solutions shared after class

Part D gives you a dataset with a strong, highly significant correlation in it and asks you to work out which of the five explanations it is. There is a right answer, and it is not "X causes Y".

TODAY IN TWELVE LINES

Recap

1. r runs −1 to +1 and measures one thing: tightness around a STRAIGHT line.
2. ⚠️ r is not a slope. Slopes of 0.09 and 90 both gave r = 0.697.
3. r^2 is the share of variance. r = 0.3 is 9%, which is almost nothing.
4. ⭐ Anscombe: four different pictures, all with r = 0.816. Plot the data.
5. ⚠️ r sees only straight lines. A perfect U-shape gives r ≈ 0.
6. Spearman = r on the ranks. Fixes curves and outliers, not U-shapes.
7. ⚠️ One point moved r from −0.14 to +0.64, at p = 4.7e-13.
8. ⚠️ r = 0.5 from n = 10 has a CI of −0.19 to +0.86. You learned nothing.
9. ⚠️ At n = 100,000, r = 0.006 is “significant” and explains 0.004%.
10. Five explanations: chance, confounding, reverse, selection — and cause.
11. ⚠️ Selection manufactured r = −0.73 between two independent variables.
12. ⚠️ df.corr() on 20 columns is 190 hypothesis tests, not one picture.

Homework

Run correlate() on three pairs of columns from your Day 21 project and report all five numbers · Find a news headline built on a correlation and work the checklist on it · Take one correlation you believe is causal and name the confounder you would need to rule out

Next class — Topic 2.10

The statistics mini project. Everything from Day 23 onwards, applied end to end on a real dataset — describe it, test something, and report it honestly enough that someone could check you.